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Simulation of Sour Gas Sweetening Using Diethanolamine (DEA) in Aspen HYSYS

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1. Introduction

The presence of acid gases, such as Hydrogen Sulfide (H₂S) and Carbon Dioxide (CO₂), in natural gas streams poses significant challenges, including severe equipment corrosion and high toxicity hazards. This project demonstrates the steady-state simulation of a gas sweetening process using a Diethanolamine (DEA) solution to efficiently remove these impurities and meet stringent pipeline specifications.

2. Thermodynamic Model & Simulation Environment

Software: Aspen HYSYS

Fluid Package: Acid Gas - Chemical Solvent

Solvent: Diethanolamine (DEA) (~28 wt% aqueous solution)

3. Process Description & Simulation Steps

3.1. Sour Gas Preparation & Absorption

The raw sour gas enters an inlet separator to remove any heavy condensates before entering the absorption column (Contactor). The absorber operates counter-currently, where the sour gas enters from the bottom (1000 psia) and the lean DEA solution enters from the top (995 psia).

The absorber efficiently strips H₂S and CO₂ from the vapor phase, producing a Sweet Gas stream at the top and a Rich DEA stream at the bottom.

3.2. Rich Amine Expansion & Flash Drum

The Rich DEA stream is routed through a control valve, dropping its pressure (Delta P = 910 psi) to flash off dissolved light hydrocarbons. The mixture then enters a Flash Separator (Flash Vap), separating the flashed vapor from the liquid rich amine.

3.3. Lean/Rich Heat Integration

To optimize energy consumption, the liquid rich amine from the flash drum passes through the tube side of a Lean/Rich Shell and Tube Heat Exchanger. It recovers heat from the hot lean amine exiting the regenerator (shell side), preheating the rich amine before it enters the regeneration column. A pressure drop of 10 psia is assumed for both the shell and tube sides.

3.4. Regeneration Column (Stripper)

The preheated rich amine enters the Regenerator (a distillation column) at tray 4. The column consists of 18 trays and operates with a full reflux condenser.

- Condenser Pressure: 27.5 psia (with a 2.5 psia pressure drop)
- Reboiler Pressure: 31.5 psia
- Reflux Ratio: 0.3
- Column Duty: 1.356×10^7 Btu/hr

Solver Configuration: To ensure rapid and stable convergence without getting stuck in lengthy calculation loops, the Fixed Damping Factor in the Solver Parameters was adjusted to 0.4, and tray efficiencies were specified for H₂S and CO₂.

3.5. Solvent Recovery & Recycle Loop

The hot lean amine exiting the regenerator's bottom passes through the Lean/Rich Heat Exchanger, cooling down to 200°F. It is then mixed with makeup water in a Mixer to compensate for any solvent losses. The stream is further cooled to 95°F using an air/water cooler (8 psia pressure drop). Finally, a Pump increases the amine pressure to match the absorber's operating pressure. A logical SET operator is utilized to dynamically link the recycle pump's discharge pressure to the absorber's bottom feed pressure (maintaining a 5 psia offset). A Recycle block is used to close the flow loop, allowing the simulation to converge completely.

4. Key Results & Conclusion

The simulation successfully achieved convergence for the closed-loop amine sweetening process. The DEA absorption system effectively reduced the acid gas concentrations in the natural gas stream. Specifically, the H₂S mole fraction in the sour gas was reduced from an initial value of 0.0172 down to a safe, trace amount of 0.0000 in the sweet gas, successfully meeting standard industrial limitations.

5. Design Data & Parameters

1. Feed Gas Specifications

Parameter	Value
Temperature	86 °F
Pressure	1000 Psia
Molar Flow	25 MMSCFD

Feed Composition:

Component	Mole Fraction
Nitrogen	0.0016
CO ₂	0.0413
H ₂ S	0.0172
Methane	0.8692
Ethane	0.0393
Propane	0.0093
i-Butane	0.0026
n-Butane	0.0029
i-Pentane	0.0014
n-Pentane	0.0012
N-Hexane	0.0018
N-Heptane	0.0072
H ₂ O	0.0050

2. Lean Solvent (DEA) Specifications

Parameter	Value
Temperature	95 °F
Pressure	995 Psia
Molar Flow	190 USGPM

3. Regeneration Column (Stripper) Specifications

Property	Value
Total Trays	18
Feed Tray	4
Condenser Pressure	27.5 Psia
Reboiler Pressure	31.5 Psia
Condenser Temp	179.6 °F
Reflux Ratio (R)	0.3
Reboiler Duty	1.356×10^7 Btu/hr

4. Equipment Performance (Heat Exchangers & Coolers)

Equipment	Inlet	Outlet	ΔP (Psia)	T _{out} (°F)
Lean/Rich HX	Regen Btms	Lean from L/R	10	200
Cooler (E-101)	DEA to Cooler	DEA to Pump	8	95

Conclusion

The simulation successfully demonstrates that the H₂S concentration can be reduced to trace levels (effectively zero) while maintaining high solvent efficiency and heat integration.

